Ope Edet

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Professional Summary

Second-year Electrical and Electronics Engineering student at Obafemi Awolowo University building systems and backend software in **C/C++** and **Python** — including a **custom Unix shell** with fork/exec/wait-based process control, a **LISP interpreter** evaluated over a custom AST, a **price-time priority limit order book and matching engine**, and the **FastAPI and PostgreSQL backend** for Netrakr, a crowdsourced campus network-quality app. **200+ problems solved** across Codeforces and LeetCode. Seeking a **backend/systems software engineering internship**.

Education

Obafemi Awolowo University (OAU) — Ile-Ife, NG
Bachelor of Science in Electrical and Electronics Engineering — *Aug. 2024 – Present*
Relevant Coursework: Embedded Systems (Arduino), Statistics

Experience

Netrakr | *Founder & Backend Lead* | *Python, FastAPI, PostgreSQL, SQLModel*

- Founder and backend lead of a **crowdsourced mobile-network quality app** that maps mobile network providers' performance across Nigerian campuses, with a **waitlist of 100+ students** from multiple universities.
- Designed and built the **REST API and relational schema** with FastAPI and SQLModel, including a **zone-assignment system** that takes multiple GPS fixes, stores accuracy, and assigns a campus zone only when the fixes agree, exposing zone names instead of raw coordinates.
- Modeled campus zones as **polygons (GeoJSON)** and managed schema changes with **Alembic migrations** in a Git team workflow with pull requests.

Projects

Custom Unix Shell | *C, Linux*

- Built a **Unix-like command shell from scratch** to gain a first-principles understanding of how operating systems manage processes and I/O.
- Implemented the core prompt loop, command parsing, and built-in commands, alongside **process creation and control using fork, exec, and wait**, plus **input/output redirection** via file descriptors.
- Currently extending the shell with **signal handling and job control** to support background processes and process groups.

LISP Interpreter | *C, GDB*

- Implemented a **Lisp interpreter in C**, building a grammar and parser with the mpc parser combinator library to construct and evaluate **abstract syntax trees (ASTs)**.
- Designed *lval* structures to represent and evaluate expressions, including **manual memory management** and Q-Expressions (unevaluated expressions) as data-manipulation primitives.

Limit Order Book / Matching Engine | *C++, CMake*

- Designing and building a **price-time priority limit order book and matching engine** in C++ to deepen practical fluency with the kind of low-latency, data-structure-heavy systems used in financial technology.
- Implemented core order book logic supporting **limit and market order matching, and order cancellation**, in a multi-file CMake structure separating interface, implementation, and tests.
- Currently extending the engine with a **concurrent producer-consumer order queue**, followed by a TCP networking layer and persistence.

Extracurriculars

1st Place, Circom — solo circuit assembly competition, SPAC IEEE OAU (Student Activities Committee, IEEE OAU SB).

President, JETS Club (Junior Engineers, Technicians and Scientists) — took full ownership of club activities and project teams, represented the school at a state-level tech fair, and grew the club's population by over **200%**.

Technical Skills

Languages: C, C++, Python, SQL
Backend: FastAPI, PostgreSQL, SQLModel, REST APIs, Alembic
Data Structures & Algorithms: hash maps, trees, priority queues, graphs; 200+ problems solved on Codeforces and LeetCode
Systems Concepts: process control, I/O redirection, memory management, concurrency, parsing (ASTs), API and database schema design
Tools: Git, Linux CLI, GDB, CMake